

# WHO IS ALAN TURING ?

Short biography about the father of computer science

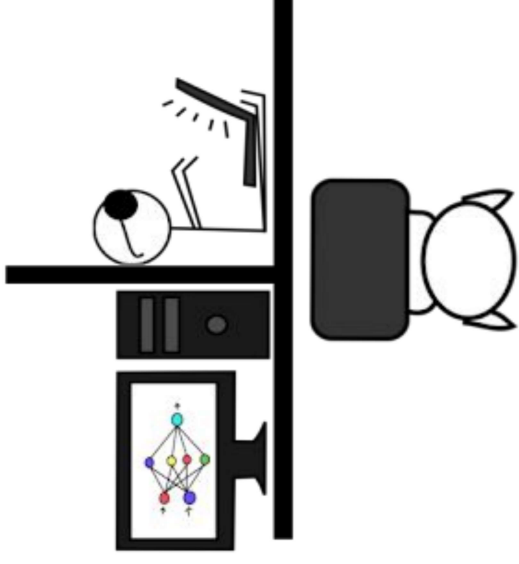
# ALAN MATHISON TURING, THE MAN...

- Born in London em 1912;
- At the 15 years old he solving complex mathematical problems, without having studied calculus;
- At the 19 years old, graduated with honors in mathematics from King's College Cambridge;
- In 1938, obtained his PhD at Princeton University, where he published several works, and then returned to England.



# THE SCIENTIST...

- Turing's main work was “On Computable Numbers” (1936), with an application to the Entscheidungsproblem;
- This revolutionary article started the fundamentals of computer science;
- Roots of the first computer appeared: a system that, alone, would carry out tasks determined by the program with which it was equipped.



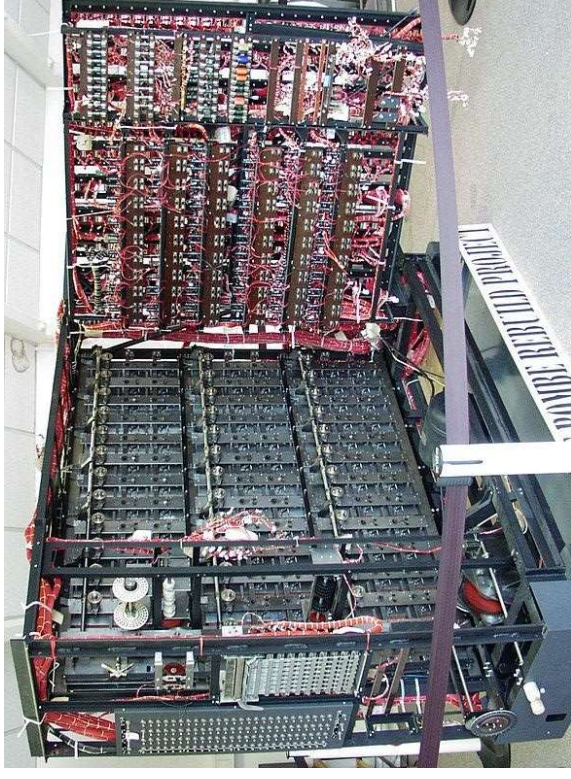
# THE WORLD WAR II

- Dependency on England for supplies transported across the North Atlantic.
- Turing worked at the British center for cryptanalysis and later headed a team called Hut 8.
- Enigma machine - Electromechanical rotor mechanism that scrambles the 26 letters of the alphabet.



# THE HERO...

- “The Bombe” – Electromechanical machine to decode the messages;
- Shortened the war in Europe by more than 2 years;
- Saved over 14 million lives;



# THE HERO'S LIFE

- 1945 – Order of the British Empire
- Professor at the University of Manchester;
- UK's National Physical Laboratory worked on the design of Automatic Computing Engine (ACE);
- Manchester 1 – The first computer with guidelines similar to those of today.
- Have interested in chemistry when he spent a period working at Bell Laboratories in the United States.



# THE INDECENCY CONVICTION

- 1952, “homosexual acts” – He refused arrest and subjected him to estrogen injections;
- Was prevented from continuing to contribute to government activities;
- Lost his security privileges granted after the War canceled.



# THE END...

- Died in 1954, believed have been suicide by ingesting cyanide, but scholars concluded caused by an accident using chemical elements in homemade experiments.
- In 2009, was an apology by the British government;
- In 2013 Turing was posthumously pardoned of the conviction for homosexual practice, by Queen Elizabeth II.



THANK YOU!